Experiment - 7

Aim:
To write a program to implement flow control at data link layer using sliding window protocol.

Code:

Sender.py

```
import time
w = int(input("Enter window size: "))
msg = input("Enter message: ")
frames = [[i, ch] for i, ch in enumerate(msg)]
base = 0
while base < len(frames):
    with open("Sender_Buffer.txt", "w") as f:
        for i in range(base, min(base + w, len(frames))):
            f.write(f"{frames[i][0]}{frames[i][1]}\n")
elif data.startswith("NACK"):
    print("NACK received. Resending...")
```

except:
print("waiting for receiver...")

Receiver.py

exp = 0
with open("Sender_Buffer.txt", "r") as f:
lines = f.readlines()

ok = True
for line in lines:
no, data = line.strip().split(";")
if int(no) != exp:
ok = False
nack = exp
break
exp += 1
with open("Receiver_Buffer.txt", "w") as f:
if ok:
f.write(f"Ack:{exp-1}")
print("Ack sent:", exp-1)

```
else:
    b.write(b"Nack: {nack}")
    print("Nack Sent :", nack)
```

I/P

Enter window size : 3

Enter Message : HELLO

O/P

Frames sent:

[0, 'H']
[1, 'E']
[2, 'L']

Ack received: 2

Frames sent:

[3, 'L']
[4, 'O']

Ack received: 4

Result -

Thus the program to implement flow control at data link layer using sliding window.